Scatter Graphs & Correlation

Exam code: 0607

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Correlation

What is correlation?

- **Correlation** describes how two quantities are related to each other
- **Positive correlation** is when one quantity **increases** and the other quantity **increases**
  - For example, as temperature increases, sales of cold drinks increase
- **Negative correlation** is when one quantity **decreases** while the other quantity **increases**
  - For example, the value of a car decreases as its age increases
- **No (zero) correlation** is where there is no apparent relationship
  - For example, the masses of snails and scores in an exam

What does the phrase "correlation does not imply causation" mean?

- If two quantities **correlate**, it does **not** mean that the first **causes** the second
- For example, each day you record the height of a sunflower and the weight of a puppy
  - As the height of the sunflower **increases**, the weight of the puppy **increases**
    - This is a **positive** correlation
  - But you **cannot claim** that:
    - If you want your puppy to weigh more, make your sunflower taller!
    - Sunflowers grow better when puppies are heavier!
  - Both quantities may be increasing due to another reason
    - In this case, **time**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Scatter graphs

pairs of data
    - For example, students' Maths grades against their Physics grades

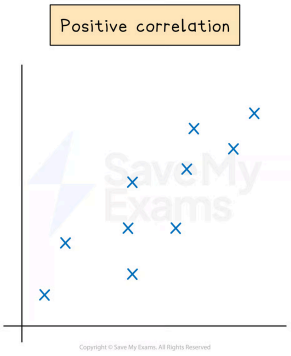
- The vertical and horizontal **axes** represent the two **quantities** being measured
- **Points** are plotted as **crosses**, ×
    - They are **not** joined up

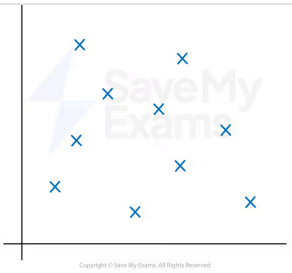
- The general **shape** formed by the points shows the type of **correlation**
    - **Positive** correlation goes from bottom left to top right
        - A positive gradient

    - **Negative** correlation goes from top left to bottom right
        - A negative gradient

    - **No (zero)** correlation looks like a cloud of points

- **Correlations** can be **weak** or **strong**
    - The **stronger** the **correlation**, the **closer to a straight line** the data points lie





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